

Helix OTA — Issues (open workable items)

Revision: 3 **Last modified:** 2026-06-19T08:30:00Z

This is the canonical open-work tracker (§11.4.15 Status, §11.4.16 Type, §11.4.54 OTA-NNN). Closed items migrate to [Fixed.md](#). The short-form companion is [Issues_Summary.md](#).

§3. [OTA-003] Emulator Tier-2 — real Android A/B (update_engine/AVB/dm-verity auto-rollback) is host-gated

Status: Operator-blocked **Type:** Task

Operator-Block-Details: - **WHAT:** Stand up the Tier-2 emulator — a real Android A/B update_engine payload-apply with AVB/dm-verity verification and auto-rollback, driven end-to-end against the control plane (per docs/design/EMULATED_DEVICE_TESTING.md). - **WHY:** Tier-2 requires Cuttlefish (cvd) on a Linux host with nested KVM. Self-resolution exhausted (§11.4.21): (a) no host CLI/virtualisation path — the current host is Apple-Silicon with applehv, which cannot run Cuttlefish or nested KVM; (b) subagent delegation cannot conjure the missing kernel virtualisation; (c) the containers submodule (podman/applehv) cannot host a KVM-backed Cuttlefish guest on this hardware; (d) no captured fallback substitutes for real update_engine partition behaviour; (e) external research confirms Cuttlefish's Linux+KVM requirement — not a tooling gap. NOT structurally impossible (§11.4.112) — host/hardware-gated only. - **UNBLOCK CONDITION:** Access to a Linux host (or CI runner) with nested-KVM enabled where Cuttlefish (cvd) boots. - **WHO:** Operator — provision the Linux+KVM host / CI runner; see docs/design/EMULATED_DEVICE_TESTING.md Tier-2.

§4. [OTA-004] Emulator Tier-3 — real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-verity on real partitions

Status: Operator-blocked **Type:** Task

Operator-Block-Details: - **WHAT:** Tier-3 validation on the physical board — real vendor HAL, U-Boot A/B slot-switch, and dm-verity over real partitions, exactly as a fielded device runs (per docs/design/EMULATED_DEVICE_TESTING.md). -

WHY: Requires the physical RK3588 / Orange Pi 5 Max hardware, which is not attached to this host. Self-resolution exhausted (§11.4.21): (a) no remote board over the available CLIs/ADB/SSH; (b) subagent delegation cannot substitute for absent silicon; (c) no repo tooling emulates real U-Boot slot-switch + on-silicon dm-verity; (d) no captured fallback reproduces real-partition AVB rollback; (e) research confirms the vendor HAL + bootloader path is board-specific. NOT structurally impossible (§11.4.112) — hardware-gated only. - **UNBLOCK CONDITION:** A physical RK3588 / Orange Pi 5 Max board reachable over ADB/SSH (or physically attached) for flashing. - **WHO:** Operator — attach / provide remote access to the board.

§8. [OTA-014] Docs Chain submodule distribution (constitution §11.4.106 Phase 6) operator-gated

Status: In progress **Type:** Task **Description:** Docs Chain engine exists at / Volumes/T7/Projects/docs_chain with Phases 1-4 done and tested. Phase 6 (git submodule distribution) is operator-gated — needs explicit go-ahead to wire as a registered submodule. Meanwhile, .docs_chain/ YAML contexts + scripts/ sync_status_docs.sh are wired and working.